



**BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH
HONOURS**

CSE3433 - SOFTWARE ARCHITECTURE

SEMESTER II: SESSION 2025/2026

Project Title:

Microservices-Based Disaster Relief Coordination Platform

GROUP 02

COURSE LECTURER:

DR. MOHAMAD NOR HASSAN

NAME	MATRIX NUMBER
MUHAMMAD ADHA BIN MUHAMMAD HAMKA	S76187
MUHAMMAD AIMAN MUKHLIS BIN KHAIRUM ANUAR	S76194
MUHAMMAD SYAKIR ZUFAYRI BIN SAARI	S74824

TABLE OF CONTENT

Architecture Style	3
Problem Description	4
System Objectives	5
Initial Functional Requirement	6
Project Members and Roles	8
System Overview	9
Functional Requirements	10
Non-Functional Requirements	12
Architecture Style Explanation	13
Service Identification	14
Architecture Diagram	15
System Context Diagram	15
Interaction Diagram	16

Architecture Style

Microservice architecture has been selected because it provides the flexibility and resilience required for a modern, evolving application by breaking it down into a suite of independent services. Unlike a traditional monolith where all features are tightly coupled, each microservice focuses on a single business task and can be developed, tested, and deployed without risking the stability of the entire system. This modularity ensures that if one service fails, the rest of the application remains functional, providing a much higher level of reliability for the end user.

This approach also embraces decentralization, distributing data management and logic across the system rather than relying on a single, massive central database. By giving each service its own dedicated data store, the architecture prevents bottlenecks and allows different teams to use the best technology for their specific needs. Furthermore, because these services operate on separate servers or cloud containers, the system can scale with extreme precision. Instead of upgrading the entire infrastructure to handle a spike in one area, you can simply allocate more server resources to the specific service under pressure, ensuring cost-efficiency and high performance.

Problem Description

In disaster situations such as floods, earthquakes or pandemics, coordination between different parties is very important but often difficult to manage. Many existing disaster management systems are still based on traditional or centralized designs, which are not flexible enough to handle sudden increases in users and data during emergencies.

One major problem is that these systems can become slow or even crash when too many people try to access them at the same time. For example, when many users report incidents or request help at the same time, the system may not respond quickly causing delays in rescue operations.

Another issue is the lack of proper communication and coordination between organizations such as government agencies, NGOs and volunteers. Information is sometimes not shared in real-time which can lead to duplicated efforts or missed areas that need urgent attention.

In addition, updating or improving a traditional system is usually difficult because all components are tightly connected. A small change in one part of the system may affect the entire system making maintenance time-consuming and risky.

Because of these limitations, there is a need for a more flexible and scalable system that can handle high demand, support real-time communication and allow different services to work independently without affecting the whole platform.

System Objectives

The objective of this project is to develop a Microservice-Based Disaster Relief Coordination Platform by focusing on three main phases which are analysis, design and development.

First, the objective is to **analyze** the current problems in existing disaster management systems. This includes identifying issues such as poor scalability, lack of real-time coordination and system failures during high demand. Through this analysis, the system requirements can be clearly understood to ensure the proposed solution addresses real-world challenges.

Next, the objective is to **design** an efficient system architecture using a microservices approach. This involves structuring the system into independent services such as incident reporting, resource management and volunteer coordination. The design should make sure that the system is scalable, flexible and capable of handling high traffic during disaster situations.

Finally, the objective is to **develop** the system based on the proposed design. This includes implementing the microservices, integrating them through APIs and ensuring that all components function correctly together. The development phase also focuses on creating a reliable and user-friendly platform that supports effective disaster response and coordination.

Overall, these objectives aim to produce a system that is well-analyzed, properly designed and successfully developed to improve disaster management operations.

Initial Functional Requirement

1. User and Authentication Module

- a. FR1 The system shall enable users (Victims, Non-Governmental Organizations, Government Agencies, and Volunteers) to create and maintain their respective accounts
- b. FR2 The system shall have a centralized API Gateway to authenticate and authorize all other services

2. Reporting Module

- a. FR3 The system shall allow victims or volunteers to submit real-time disaster reports, including location (GPS), type of emergency, and photos.
- b. FR4 The system shall categorize and prioritize incidents based on severity levels (e.g., High, Medium, Low).
- c. FR5 The system shall provide an automated notification to nearby responders when a high-priority incident is logged

3. Logistics Management Module

- a. FR6 The system shall permit NGOs and government institutions to register their resources, such as food, medication, and equipment that can be used in rescue operations in various relief centers in real time.
- b. FR7 The system shall automate the handling of requests made for resources by incidents. In addition, it shall automatically record stock updates on successful delivery.

4. Volunteer Management Module

- a. FR8 The system shall allow volunteers to register, update their profiles, and specify their skills (e.g., medical, rescue, logistics).
- b. FR9 The system shall match volunteers to incidents automatically based on location, availability, and skillset.
- c. FR10 The system shall allow volunteers to accept or reject assigned tasks in real time.

5. Communication Module

- a. FR11 The system shall provide real-time messaging between victims, responders, NGOs, and government agencies.
- b. FR12 The system shall support broadcast announcements to all users within a specific geographic area.
- c. FR13 The system shall log communication history for monitoring and auditing purposes.

6. Notification Module

- a. FR14 The system shall send notifications via multiple channels (mobile push notifications, SMS, or email).
- b. FR15 The system shall allow users to customize notification preferences.
- c. FR16 The system shall trigger alerts for critical events such as resource shortages or high-risk incidents.

Project Members and Roles

Name	Role	description
MUHAMMAD ADHA BIN MUHAMMAD HAMKA	Project Manager	Define project scope and timeline, assign tasks to team members, monitor progress, ensure the project is completed on time.
MUHAMMAD AIMAN MUKHLIS BIN KHAIRUM ANUAR	Database Admin	Responsible for designing, implementing, and maintaining the database system. Manages user access, ensures data security, and performs backups and recovery.
MUHAMMAD SYAKIR ZUFAYRI BIN SAARI	System Architect	Decide how services are divided (e.g., user service, resource service), design communication between services, choose technologies and frameworks, Ensure scalability and reliability.